

Chapter 2: Review of Probability and Statistics

Multiple Choice

1. Continuous random variables

Which of the following random variables is a *continuous* random variable?

- (a) The time (in minutes) until the next customer enters a shop.
- (b) The payment method $\in \{\text{cash}, \text{card}\}$ of the next customer.
- (c) The number of defective items in a delivery of 100 items.
- (d) The result of rolling a die.
- (e) The number of customers who enter a shop within one hour.

2. Properties of a density function

Let f_X be the density function of a continuous random variable X . Which statement is correct?

- (a) $\int_{-\infty}^{\infty} f_X(x) dx$ can be any positive number.
- (b) $P(a \leq X \leq b) = f_X(b) - f_X(a)$ for all $a \leq b$.
- (c) $f_X(x)$ can take negative values, as long as $\int_{-\infty}^{\infty} f_X(x) dx = 1$.
- (d) $f_X(x) = P(X = x)$ for all x .
- (e) $f_X(x)$ can take values larger than 1.

3. Probabilities from a probability mass function

The number of complaints W per day has the probability mass function

w_k	0	1	2	3	4
$p_W(w_k)$	0.05	0.20	0.35	0.25	0.15

What is $P(W \geq 2)$?

- (a) 0.60
- (b) 0.75
- (c) 0.95
- (d) 0.35
- (e) 0.40

4. Expected value of a discrete random variable

The number of items W that a customer buys has the probability mass function

w_k	1	2	3	4
$p_W(w_k)$	0.2	0.4	0.3	0.1

What is $E(W)$?

- (a) 2.30
- (b) 6.10
- (c) 0.58
- (d) 2.00
- (e) 2.50

5. Probabilities from a cdf

The continuous random variable X has the cumulative distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ (x/5)^2 & 0 \leq x \leq 5 \\ 1 & x > 5. \end{cases}$$

What is $P(1 < X \leq 3)$?

- (a) 0.32
- (b) 0.40
- (c) 0.96
- (d) 0.16
- (e) 0.36

6. Standard deviation of a linear transformation

The monthly sales X (in units) of a product have standard deviation $SD(X) = 80$. The monthly profit (in EUR) is $\Pi = 15X - 1000$. What is $SD(\Pi)$?

- (a) 1,440,000
- (b) 1,200
- (c) 18,000
- (d) 200
- (e) 80

7. Variance of a linear transformation

Let X be a random variable with $\text{Var}(X) = 5$. What is $\text{Var}(-2X + 10)$?

- (a) -10
- (b) 30
- (c) 10
- (d) 0
- (e) 20

8. Sample variance

We observe $n = 4$ values: $X_{1,\text{obs}}, \dots, X_{4,\text{obs}} = 8, 7, 1, 4$. What is the sample variance $\hat{\sigma}_{X,\text{obs}}^2$?

- (a) 2.50
- (b) 30.00
- (c) 10.00
- (d) 7.50
- (e) 3.16

9. Correlation

Two random variables X and Y have $\text{Var}(X) = 16$, $\text{Var}(Y) = 25$ and $\text{Cov}(X, Y) = 15.00$. What is $\text{Cor}(X, Y)$?

- (a) 0.366
- (b) 0.600

- (c) 0.038
- (d) 0.750
- (e) 0.938

10. Covariance, correlation and units of measurement

For a sample of stores, the advertising expenditures X are measured in 1000 EUR and the sales Y in 1000 units. The sample covariance is $\widehat{\text{Cov}}_{\text{obs}}(X, Y) = 4$ and the sample correlation is $\widehat{\text{Cor}}_{\text{obs}}(X, Y) = 0.8$. Now, the advertising expenditures are measured in EUR instead (i.e., all X -values are multiplied by 1000). Which statement is correct?

- (a) The covariance becomes 4,000,000, and the correlation stays 0.8.
- (b) The covariance becomes 4000, and the correlation becomes 0.0008.
- (c) The covariance becomes 4000, and the correlation stays 0.8.
- (d) The covariance becomes 4000, and the correlation becomes 800.
- (e) The covariance stays 4, and the correlation stays 0.8.

11. Conditional mean

The joint probability mass function $p_{Y,X}(y, x) = P(Y = y, X = x)$ of two discrete random variables $Y \in \{0, 1, 2\}$ and $X \in \{0, 1\}$ is given by

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.10	0.20	0.10
$X = 1$	0.30	0.20	0.10

What is the conditional mean $E(Y \mid X = 1)$?

- (a) 0.667
- (b) 0.600
- (c) 0.800
- (d) 1.000
- (e) 0.400

12. Independence

Let Y and X be two continuous random variables with joint density $f_{Y,X}$, marginal densities f_Y and f_X , and conditional density $f_{Y|X}$. Which statement is correct?

- (a) If $\text{Cor}(X, Y) = 0$, then Y and X are always independent.
- (b) Y and X are independent if $E(Y) = E(X)$.
- (c) If Y and X are independent, then $f_{Y,X}(y, x) = f_Y(y) + f_X(x)$.
- (d) If Y and X are independent, then $\text{Cor}(X, Y) = 1$.
- (e) If Y and X are independent, then $f_{Y|X}(y \mid x) = f_Y(y)$ for all y and x .

13. Normal probabilities

Let $X \sim \mathcal{N}(100, 10^2)$ and let Φ denote the cdf of the standard normal distribution, with $\Phi(2.0) = 0.9772$. What is $P(X > 120)$?

- (a) 0.9545
- (b) 0.0455
- (c) 0.0228
- (d) 0.9772

(e) 0.5000

14. Chi-squared distribution

Let $Z_1, \dots, Z_3$ be independent standard normal random variables and $W = Z_1^2 + \dots + Z_3^2$. What is $\text{Var}(W)$?

(a) 9.00

(b) 2.00

(c) 3.00

(d) 6.00

(e) 2.45

15. F-distribution

Let $W_1 \sim \chi_3^2$ and $W_2 \sim \chi_{10}^2$ be independent, and let Z_1, Z_2 be independent standard normal random variables. Which random variable has an $F_{3,10}$ distribution?

(a) $\frac{Z_1}{Z_2}$

(b) $\frac{W_2/10}{W_1/3}$

(c) $\frac{W_1}{W_2}$

(d) $\frac{W_1 + W_2}{13}$

(e) $\frac{W_1/3}{W_2/10}$

16. Regression function of a DGP

The data are generated by $Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i$ with $\beta_1 = 40$, $\beta_2 = 2$, and $\varepsilon_i \sim \mathcal{N}(0, 8^2)$ independent of X_i . What is $E(Y_i | X_i = 10)$?

(a) 42.0

(b) 20.0

(c) 60.0

(d) 68.0

(e) 402.0

17. Law of Large Numbers

Let $X_1, \dots, X_n$ be an i.i.d. random sample with mean $E(X_i) = \mu$. What does the Law of Large Numbers say?

(a) For $n \geq 30$, the sample mean $\bar{X}_{\text{obs}}$ is exactly equal to μ .

(b) For large n , the observations $X_{1,\text{obs}}, \dots, X_{n,\text{obs}}$ are approximately normally distributed.

(c) The sample mean $\bar{X}_{\text{obs}}$ gets close to μ as the sample size n grows.

(d) The variance of each X_i decreases as n grows.

(e) For large n , every single observation $X_{i,\text{obs}}$ is close to μ .

18. Association and causation

In a sample of primary school children, shoe size and reading skills are strongly positively correlated. Which statement is correct?

- (a) A strong correlation always implies a causal effect in at least one direction.
- (b) Since the correlation is strong, the relationship cannot be explained by a third variable.
- (c) The correlation is most likely driven by a confounder: older children have larger feet *and* read better.
- (d) Better reading skills cause larger feet.
- (e) Larger shoes cause better reading skills.